

Coral Reefs Under Environmental Change

A Teacher Implementation Guide for 9th-Grade Biology

Aligned to the Texas Essential Knowledge and Skills (TEKS) for Biology, 19 TAC §112.42

Developed and implemented by Dr. Alexandra Good (Texas A&M University–Corpus Christi) in collaboration with the Texas State Aquarium Education Department and the TAMU-CC NSF Noyce Scholars program. Piloted with 9th-grade Biology students at West Oso High School, Corpus Christi, Texas.

This guide gives any Texas Biology teacher everything needed to run a four-phase unit on coral reefs and environmental change. Each lesson includes learning objectives, the specific Biology TEKS it addresses, a materials list, advance preparation, a step-by-step procedure, and the matching student handout. You can teach the full unit in roughly seven to nine class periods, or pull individual lessons into an existing ecology, energy, or human-impact module.

The unit moves from foundational content delivery to hands-on science and real-world stewardship action:

- **Phase 1: Classroom foundation** - two days of coral reef biology and ecology.
- **Phase 2: Virtual field trip** - an immersive visit to a Hawai‘i research station, a documentary, and scientist interviews.
- **Phase 3: Hands-on lab** - measuring coral growth under ocean-acidification conditions and an oyster shell dissolution demonstration.
- **Phase 4: Local action** - a marine-debris art activity and a beach cleanup.

All student handouts, slide decks, readings, and data files referenced here are provided in the accompanying “For Teachers” materials folder.

Curriculum Overview

Lesson	Format	Time	Core Biology TEKS
1. Coral Reef Biology Lecture	Lecture & discussion	2 classes	1.A, 1.G, 3.A, 3.C, 11.A, 13.A, 13.B
2. Virtual Field Trip: Aquavision & Chasing Coral Documentary	Virtual / media	2–3 classes	1.A, 4.A, 4.B, 13.D
3. Virtual Field Trip: Meet a Scientist	Virtual interviews	1 class	4.B, 4.C
4. Ocean Acidification Lab	Hands-on lab & data analysis	2 classes	1.B, 1.C, 1.E, 2.B, 2.C, 3.A, 3.B, 13.C
5. Trash in the Ocean: Art Activity & Beach Cleanup	Art activity & field trip	2 days	1.A, 3.A, 3.C, 13.D

Biology TEKS addressed

The expectations below are drawn from 19 Texas Administrative Code §112.42 (Biology, adopted 2020, implemented beginning 2023–2024). Codes use the subsection letter and number as cited in the rule.

Scientific and engineering practices

TEKS	Student expectation	Used in
1.A	Ask questions and define problems based on observations or information from text, phenomena, models, or investigations.	Lessons 1, 2, 4, 5
1.B	Apply scientific practices to plan and conduct descriptive, comparative, and experimental investigations and use engineering practices to design solutions to problems.	Lesson 4
1.C	Use appropriate safety equipment and practices during laboratory, classroom, and field investigations.	Lessons 4
1.E	Collect quantitative data using the International System of Units (SI) and qualitative data as evidence.	Lesson 4
1.G	Develop and use models to represent phenomena, systems, processes, or solutions.	Lesson 1
2.B	Analyze data by identifying significant statistical features, patterns, sources of error, and limitations.	Lesson 4
2.C	Use mathematical calculations to assess quantitative relationships in data.	Lesson 4
3.A	Develop explanations and propose solutions supported by data and models and consistent with scientific ideas, principles, and theories.	Lessons 1, 4, 5
3.B	Communicate explanations and solutions individually and collaboratively in a variety of settings and formats.	Lesson 4
3.C	Engage respectfully in scientific argumentation using applied scientific explanations and empirical evidence.	Lessons 1 and 5
4.A	Analyze, evaluate, and critique scientific explanations and claims using empirical evidence and scientific argumentation.	Lesson 2
4.B	Relate the impact of past and current research on scientific thought and society, including contributions of diverse scientists.	Lessons 2, 3
4.C	Research and explore resources such as museums, professional organizations, online platforms, and mentors employed in a STEM field to investigate STEM careers.	Lesson 3

Science concepts

TEKS	Student expectation	Used in
11.A	Explain how matter is conserved and energy is transferred during photosynthesis and cellular respiration using models, including the chemical equations for these processes.	Lesson 1
13.A	Investigate and evaluate how ecological relationships, including predation, parasitism, commensalism, mutualism, and competition, influence ecosystem stability.	Lesson 1
13.B	Analyze how ecosystem stability is affected by disruptions to the cycling of matter and flow of energy through trophic levels using models.	Lesson 1
13.C	Explain the significance of the carbon and nitrogen cycles to ecosystem stability and analyze the consequences of disrupting these cycles.	Lesson 4
13.D	Explain how environmental change, including change due to human activity, affects biodiversity and analyze how changes in biodiversity impact ecosystem stability.	Lessons 1, 2, 5

Lesson Plans

Lesson 1. Coral Reef Biology Lecture

Format: In-class lecture and discussion.

Time: Two class periods (about 50 minutes each).

Day 1

Learning objectives

- Define a coral as a living animal and support the claim with evidence and reasoning.
- Describe what corals need to survive and explain the mutualism between coral polyps and zooxanthellae.
- Distinguish coral feeding strategies (heterotrophic and autotrophic) and growth forms (colonial and solitary).
- Identify and give reef examples of mutualism, commensalism, parasitism, predation, and competition (direct and indirect).

Biology TEKS addressed

TEKS	Student expectation (Biology §112.42)	How this lesson addresses it
1.A	Ask questions and define problems based on observations, phenomena, or models.	The “What is a coral?” claim-evidence-reasoning warm-up frames the unit’s driving questions.
1.G	Develop and use models to represent phenomena and processes.	Students use Hawaiian Ko‘a identification cards and sketch Darwin’s model of reef formation.
3.A	Develop explanations supported by data and models, consistent with scientific ideas.	Students build a claim-evidence-reasoning explanation that corals are living animals.
11.A	Explain how matter is conserved and energy is transferred during photosynthesis and cellular respiration.	The coral–zooxanthellae mutualism is used to connect photosynthesis to the energy and matter corals receive.
13.A	Investigate and evaluate how ecological relationships, including predation, parasitism, commensalism, mutualism, and competition, influence ecosystem stability.	Students define and find reef examples of each ecological relationship and discuss how they keep a reef stable.

Materials

- ‘Coral Reef Biology Day 1’ slide deck and access to YouTube videos.
- ‘Coral Biology Day 1’ student worksheet (one per student).
- Projector with audio.

Procedure

1. Warm-up: students list things humans need to survive, then attempt “What is a coral?” using a claim, evidence, and reasoning frame.

2. Introduce key vocabulary: polyp, colonial, solitary, autotrophic and heterotrophic feeding, commensalism, competition (indirect and direct), parasitism, mutualism, predation, zooxanthellae, calcification, coral bleaching, and reef.
3. Present coral anatomy and the coral–zooxanthellae mutualism; connect photosynthesis to the energy and nutrients corals gain.
4. Walk through coral reproduction, growth forms, and competition using slides and short videos.
5. Ko‘a card activity: students record observations of the shape and color of two coral specimens.
6. Close with a model sketch of Darwin’s proposed reef formation (fringing reef to barrier reef to atoll).

Assessment

Completed ‘Coral Biology Day 1’ worksheet and exit discussion.

Day 2

Learning objectives

- Explain why coral reefs are biodiversity hotspots even though they cover less than 1% of the ocean floor.
- Describe ocean trophic structure and how energy flows through reef food webs.
- Explain the role of reefs in the carbon cycle and how climate change disrupts carbon balance.
- Analyze how environmental change and land-based pollution threaten reef stability and biodiversity, and propose actions.

Biology TEKS addressed

TEKS	Student expectation (Biology §112.42)	How this lesson addresses it
3.C	Engage respectfully in scientific argumentation using applied explanations and empirical evidence.	Students argue, with evidence, for the best way to protect the ocean.
11.A	Explain how matter is conserved and energy is transferred during photosynthesis and cellular respiration.	Calcification and photosynthesis are revisited in the context of reef carbon flow.
13.B	Analyze how ecosystem stability is affected by disruptions to the cycling of matter and flow of energy through trophic levels using models.	Students trace energy through ocean food webs and analyze how disruptions destabilize the reef.
13.D	Explain how environmental change, including human activity, affects biodiversity and analyze impacts on ecosystem stability.	Students connect climate change and land-based pollution to reef decline and biodiversity loss.

Materials

- ‘Coral Reef Biology Day 2’ slide deck and access to YouTube videos
- ‘Coral Biology Day 2’ student worksheet (one per student).
- Projector with audio.

Procedure

1. Warm-up recap video; students write three takeaways from Day 1 and any remaining questions.
2. Reef ecology: biotic and abiotic factors, biodiversity, and ecosystem services.

3. Energy flow: ocean food webs and trophic structure; reefs as the “rainforests of the sea.”
4. Biogeochemical cycles: the reef carbon cycle, carbon sinks, and the nitrogen cycle.
5. Threats: how climate change disrupts the carbon cycle, why reefs are dying, and land-based pollution sources.
6. Take-action discussion: students write an evidence-based opinion on the best way to protect ocean ecosystems.

Assessment

Completed ‘Coral Biology Day 2’ worksheet and the written opinion paragraph.

Lesson 2. Virtual Field Trip: Aquavision Expedition & Chasing Coral Documentary

Format: Virtual and media-based.

Time: Two or three classes.

This lesson takes students inside a working reef research station without leaving the classroom. The Texas State Aquarium Aquavision program is a virtual coral reef expedition filmed at the Hawai‘i Institute of Marine Biology (HIMB) on Moku o Lo‘e (Coconut Island) in Kāne‘ohe Bay, O‘ahu, one of the most studied reef systems in the world. The series follows Dr. Keisha Bahr, coral reef biologist and Associate Professor of Marine Biology at Texas A&M University–Corpus Christi, who also chairs Coral Reef and Ocean Health at the Harte Research Institute. The program is organized into four sections: (1) What Does a Scientist Look Like, (2) Coral Biology, (3) Coral Ecology, and (4) Coral Reef Impacts and Solutions.

Learning objectives

- Explore a working coral reef research station and virtually observe real experiments.
- Describe how scientists study coral responses to ocean warming and acidification.
- Analyze evidence of coral bleaching and connect it to environmental change.

Biology TEKS addressed

TEKS	Student expectation (Biology §112.42)	How this lesson addresses it
1.A	Ask questions and define problems based on phenomena and investigations.	The Chasing Coral worksheet prompts students to generate and answer questions about reef loss.
4.A	Analyze, evaluate, and critique scientific explanations and claims using empirical evidence.	Students evaluate the evidence presented in the documentary about warming oceans and reef decline.
4.B	Relate the impact of past and current research on scientific thought and society, including diverse scientists.	Students see how active research at HIMB shapes understanding of reef health.
13.D	Explain how environmental change, including human activity, affects biodiversity and analyze impacts on ecosystem stability.	Aquavision and Chasing Coral show bleaching and reef loss as consequences of environmental change.

Materials

- [Texas State Aquarium Aquavision program](#) (virtual reef expedition).
- [Chasing Coral](#) documentary (Netflix).
- ‘Chasing Coral’ student worksheet (one per student).

Procedure

1. Introduce the virtual expedition and the research site (HIMB, Kāne‘ohe Bay, Dr. Keisha Bahr).
2. View the Aquavision sections in order: What Does a Scientist Look Like; Coral Biology; Coral Ecology; Coral Reef Impacts and Solutions.
3. Screen Chasing Coral; students complete the documentary worksheet as they watch.
4. Debrief Discussion: how bleaching happens, and what the evidence shows about reef futures.

Assessment

Completed ‘Chasing Coral’ worksheet and class debrief.

Lesson 3. Meet a Scientist: Exploring STEM Careers

Format: Virtual interviews, student-driven.

Time: One class period (flexible; pairs with Lesson 2).

Students choose from a set of pre-recorded interviews with marine and environmental scientists. Each video comes with a short biography so students can pick interviews that match their interests. Every featured professional represents a different career pathway and answers the same questions about their background, journey, current research, and advice for aspiring scientists. The featured scientists come from different personal, educational, and professional backgrounds, so students can find someone whose path resembles their own. This student-driven format lets a wide range of learners see themselves in science.

Learning objectives

- Investigate marine and environmental science career pathways.
- Compare the backgrounds and journeys of diverse scientists.
- Reflect on personal interest in and confidence towards pursuing a STEM career.

Biology TEKS addressed

TEKS	Student expectation (Biology §112.42(c))	How this lesson addresses it
4.B	Relate the impact of current research on scientific thought and society, including contributions of diverse scientists.	Students compare the research and contributions of scientists from different backgrounds.
4.C	Research and explore resources and mentors employed in a STEM field to investigate STEM careers.	Students explore real marine and environmental STEM careers through diverse scientist interviews.

Materials

- [Scientist interview playlist](#) (YouTube).
- ‘Meet a Scientist’ student worksheet (one per student).

Procedure

1. Explain the format: pre-recorded interviews chosen by interest using the biographies.
2. Students view at least three or four interviews and complete the worksheet (what they learned, what was interesting, and a question they would ask).
3. Optional: students share standout careers and reflect on their own interests.

Assessment and handout

Completed ‘Meet a Scientist’ worksheet.

Lesson 4. Ocean Acidification Lab: Measuring Coral Growth and Shell Dissolution

Format: Hands-on lab and data analysis.

Time: Two class periods (for an overnight oyster-shell soak).

Students work with preserved, non-living coral skeletons from an NSF-funded experiment (OCE-2049406) conducted at the Hawai'i Institute of Marine Biology and processed at Texas A&M University–Corpus Christi. Before the experiment, each fragment was stained with alizarin dye, so students could measure linear extension growth as the distance from the dye line to the skeletal edge. They compare two species, *Montipora capitata* (MC) and *Pocillopora acuta* (PA), under a control treatment and an ocean-acidification (OA) treatment. In the absence of physical coral skeletons, teachers can request images of coral from the Coral Reef and Ocean Health group at the Harte Research Institute (Texas A&M University–Corpus Christi). Students can measure linear extension from the scaled photographs in place of handling the skeletons. A supplemental oyster-shell demonstration shows calcium carbonate dissolving in acid, connecting the lab to local coastal life.

Learning objectives

- Plan and conduct a comparative investigation of coral growth under control and ocean-acidification conditions.
- Collect quantitative data in SI units using calipers and a balance.
- Calculate average growth, graph results, and identify patterns and sources of error.
- Demonstrate calcium carbonate dissolution and connect it to ocean acidification chemistry.
- Communicate conclusions in a written lab report.

Biology TEKS addressed

TEKS	Student expectation (Biology §112.42)	How this lesson addresses it
1.A	Ask questions and define problems based on phenomena and investigations.	Students ask questions and write hypotheses about coral growth in OA
1.B	Plan and conduct descriptive, comparative, and experimental investigations.	Students run a comparative coral-growth investigation across species and treatments.
1.C	Use appropriate safety equipment and practices during investigations.	Students follow safety practices when handling vinegar, specimens, and tools.
1.E	Collect quantitative data using SI units and qualitative data as evidence.	Students measure growth in millimeters and mass in grams.
2.B	Analyze data by identifying significant statistical features, patterns, sources of error, and limitations.	Students compare treatments and species and discuss measurement error.
2.C	Use mathematical calculations to assess quantitative relationships in data.	Students compute average growth and change in shell mass.
3.A	Develop explanations and propose solutions supported by data and models.	Students draw conclusions about acidification from their own measurements.
3.B	Communicate explanations and solutions in a variety of formats.	Students submit a written lab report with tables and graphs.

TEKS	Student expectation (Biology §112.42)	How this lesson addresses it
13.C	Explain the significance of the carbon cycle and analyze consequences of disrupting it.	The lab links rising CO ₂ and ocean acidification to calcifying organisms.

Materials

- Preserved, alizarin-stained coral skeletons: *Montipora capitata* and *Pocillopora acuta* (control and OA sets).
- Calipers and metric rulers; a balance or scale.
- Oyster shells, white vinegar, water, and pH test strips, or a pH meter.
- ‘Ocean Acidification Lab’ slide deck; ‘Ocean Acidification Reading’; ‘OA Lab Report Worksheet’.
- OA Lab Data spreadsheet (to plot data or use sample data for reference without physical specimens).

Preparation and safety

- Vinegar is a mild acid: provide eye protection, avoid contact with eyes and skin, and ventilate the room.
- Set up specimen stations with calipers; review how to read calipers to the nearest tenth of a millimeter.

Procedure: Day 1

1. Pre-lab reading and discussion; students write the ocean-acidification and calcium-carbonate equations and state research questions and hypotheses.
2. Safety briefing.
3. Coral measurement: in lab groups, measure linear extension (dye line to skeletal edge) three times per specimen for both species under control and OA treatments.
4. Oyster setup: record the pH of water and vinegar, weigh each oyster shell (Day 1 mass), and submerge shells in vinegar overnight.

Procedure: Day 2

1. Re-weigh oyster shells (Day 2 mass) and calculate the change in mass.
2. Calculate average coral growth for each species and treatment, then graph the results.
3. Analyze: compare species and treatments and identify patterns, anomalies, and sources of error.
4. Write the lab report: research question, hypothesis, data, discussion, and conclusion.

Assessment and handouts

Completed lab report with data tables and graphs.

Lesson 5. Trash in the Ocean: Art Activity & Beach Cleanup

Format: In-class art activity and field trip.

Time: Two days.

The unit closes by turning global concern into local action. Students learn how marine debris harms coastal and marine life, build sculptures from cleaned debris to visualize and reframe waste, and take part in a beach or community cleanup. The pilot took place on North Beach in Corpus Christi, and the debris-art activity was led by a local Girl Scout.

Learning objectives

- Explain how marine debris affects coastal and marine ecosystems and the organisms within them.
- Connect human activity to ecosystem stability and biodiversity loss.
- Take local stewardship action by participating in a beach or community cleanup.
- Communicate environmental responsibility creatively through debris art.

Biology TEKS addressed

TEKS	Student expectation (Biology §112.42)	How this lesson addresses it
1.A	Ask questions and define problems based on observations.	Students define the local pollution problem they observe and act on.
3.A	Develop explanations and propose solutions supported by evidence.	Students propose and carry out a stewardship solution (the cleanup).
3.C	Engage respectfully in scientific argumentation using evidence.	Students discuss and defend the value of local action for global problems.
13.D	Explain how environmental change due to human activity affects biodiversity and analyze impacts on ecosystem stability.	Students link marine debris directly to harm to coastal organisms and habitats.

Materials

- ‘Trash in the Ocean Lesson Plan’ slide deck.
- Cleaned marine or community debris, with art supplies (hot glue, glue dots, markers, etc).
- Tablecloths or a tarp, gloves, and trash bags.
- For the cleanup field trip: gloves, trash bags, sun protection, and water for every student.

Preparation and safety

- Collecting debris: students can gather around the school, bring clean items from home, or collect during the cleanup.
- Cleaning debris: wear gloves; remove unsafe items (weapons, rusty metal, anything with barnacles, paper products); wash; soak hard-to-clean pieces in bleach for 10 minutes; rinse; and air-dry.
- Field trip: follow your school’s field-trip safety and supervision policies.

Procedure

1. Classroom day: present the marine pollution lesson and discuss the sources and impacts of ocean debris.
2. In groups, students build sculptures from cleaned debris to reframe waste and stewardship.
3. Field trip day: conduct a beach or community cleanup (the pilot used North Beach, Corpus Christi).
4. Debrief: connect the local action back to global reef and ocean health.

Order is flexible: students can do the cleanup first and build from what they collect, or do the classroom day first and close with the cleanup.

Assessment and handout

Participating in the cleanup and art activity, followed by a reflection discussion. The ‘Last Day Interview Questions’ could be used as an end-of-program reflection.

Program Evaluation (Optional)

The pilot measured changes in student attitudes, self-efficacy, STEM career interest, and environmental and coral knowledge using the survey across four timepoints (a baseline before the unit, one after the classroom lecture, one after the virtual and hands-on activities, and one after the field trip), along with baseline and exit interviews. Teachers who want to gauge impact can administer the provided survey at the same intervals.

- ‘First Day Interview Questions’ and ‘Last Day Interview Questions’ (baseline and exit reflection).
- Qualtrics Survey (student attitude, career, environment, and coral knowledge).

Resources and Media Links

All videos below are referenced in the lessons. Confirm access and playback before class.

Video	Link
Chasing Coral (documentary)	https://www.netflix.com/title/80168188
Scientist Interviews	https://www.youtube.com/playlist?list=PLvvC_gClROZnNgnEdJbf97jvjNzWrl4sY
Texas State Aquarium Aquavision	https://www.texasstateaquarium.org/conservation-series/conservation-series-coral/
The Coral and the Algae	https://www.youtube.com/watch?v=JENUAv0w8Q4
Coral Spawning Event	https://www.youtube.com/watch?v=eO_2JJynlOA
Larvae Settlement	https://www.youtube.com/watch?v=e7H5FZykTVU
Cleaner Wrasse	https://www.youtube.com/watch?v=9OoNNqgqeF4
Gobies & Shrimp	https://www.youtube.com/watch?v=GGgWfs65I7Q
Clownfish & Anemone	https://www.youtube.com/shorts/Hq15brb8Ilw
Coral Reefs 101	https://www.youtube.com/watch?v=ZiULxLLP32s
Why Are Coral Reefs Dying	https://www.youtube.com/watch?v=UWiorwikCMQ
What Is Ocean Acidification	https://www.youtube.com/watch?v=gZGj0BbDT38
Why Are Coral Reefs So Important	https://www.youtube.com/watch?v=s7t95oMzY-M

Sourcing, Attributions, and Safety

Coral specimens

The coral skeletons used in the lab are non-living and come from an NSF-funded research project (OCE-2049406) that studied the effects of varied ocean chemistry on coral growth. The work was conducted at the Hawai‘i Institute of Marine Biology, and all samples used in the classroom were processed at Texas A&M University–Corpus Christi after the experiment ended. Fragments were stained with alizarin dye before exposure, allowing students to measure linear extension growth.

Program partners

The virtual field trip uses the Texas State Aquarium Aquavision program, filmed at the Hawai‘i Institute of Marine Biology and led by Dr. Keisha Bahr (Texas A&M University–Corpus Christi; Harte Research Institute). The curriculum was developed by Dr. Alexandra Good with the Texas State Aquarium Education Department and the TAMU-CC NSF Noyce Scholars program, and piloted at West Oso High School in Corpus Christi, Texas.

General safety

- Lab: eye protection with vinegar; careful handling of calipers and specimens; follow TEA-approved safety standards.
- Debris handling: gloves at all times; remove and discard unsafe items; bleach-clean as described in Lesson 5.
- Field trip: follow district supervision, transportation, and sun-and-water-safety policies.

TEKS citation

Standards are cited from 19 Texas Administrative Code, Chapter 112, Subchapter C, §112.42 (Biology), adopted 2020 and implemented beginning in the 2023–2024 school year. Confirm the current wording against the Texas Education Agency text before formal documentation.